Volume I · Chapter 5 — Medical NLP Basics (regex → embeddings) · Theory

> Source: split from `med-tech-curriculum-V1.md` (V1). From this split onward this chapter is maintained **independently** here. See `Volume-I-Chapter-5-context.md` for the AI interaction log.

Prerequisites: Ch 1 (Python/Pandas).

Learning outcomes — the student can:

- Process clinical free text with **classical NLP** (tokenization, normalization, TF-IDF).
- Build **regex/rule-based extractors** for clinical concepts and detect **negation**.
- Use **word embeddings** (general and biomedical) and build a simple clinical **text classifier**.
- Articulate the **limits of classical NLP** (context, negation) that motivate transformers (bridge to Volume II).

Topics (theory) — 10 h:

#	Topic	h
5.1	Clinical text: characteristics & challenges (abbreviations, negation, jargon)	1
5.2	Preprocessing: tokenization, normalization, stopwords, stemming/lemmatizati	2

#	Topic	h
	on	
5.3	Regular expressions for clinical information extraction	2
5.4	Representations: bag-of-words, TF-IDF	1
5.5	Word embeddings: word2vec/GloVe; biomedical embeddings	2
5.6	Simple text classification, basic NER & negation detection (e.g., NegEx)	1
5.7	Limits of classical NLP → why transformers (bridge to Volume II)	1

Datasets/tools: synthetic/de-identified clinical notes; Python `re`, scikit-learn, spaCy/NLTK, gensim or pretrained biomedical embeddings.

Assessment: regex extractor deliverable — rubric (**40%**); text-classification lab (**40%**); quiz (**20%**).

Key decisions: rule-based vs. statistical NLP; when regex suffices vs. needs ML; **general vs. biomedical** embeddings; handling **negation/context**.

References: plan §13 → *Transformers & the Hugging Face ecosystem* (bridge); *Programming & data science*.

Hours: Theory **10** + Lab **10** = **20**.